

Finitely Repeated Prisoners' Dilemma: SPNE; and NE Outcome

Consider this prisoners' dilemma stage game, G_{PD} , and suppose it is played $T \geq 2$ times, without

		Player 2	
		C	D
Player 1	C	2, 2	-1, 3
	D	3, -1	0, 0

discounting, and denote the repeated game by G_{PD}^T (I suppress the discount factor). The point here is that there is a unique SPNE and a unique NE *outcome* of G_{PD}^T . In particular,

Facts. *The unique SPNE is for each player to “choose D after every history;” the unique NE outcome is “(D, D) at each date” (though with $T \geq 2$, there is more than one NE).*

SPNE. Fix $T \geq 2$. I will use induction on the number of dates τ until the end of the game. Let $\tau = 0$, and consider the subgame starting at any $h_{T-\tau-1} = h_{T-1} \in H_{T-1}$. The unique NE of this subgame is (D, D) . Now let $\tau \geq 1$, consider the subgame starting at any $h_{T-\tau-1} \in H_{T-\tau-1}$, and suppose that for $i = 1, 2$, $f_i^t(h_{t-1}) = D$ for every $h_{t-1} \in H_{t-1}$, and $t \geq T - \tau + 1$. The outcome from $t = T - \tau + 1$ to T is therefore “ (D, D) at every date $t \geq T - \tau + 1$, no matter what the players choose at $h_{T-\tau-1}$.” Player i 's utility starting at $h_{T-\tau-1}$ is

$$\pi_i(a_1, a_2) + \tau \times \pi_i(D, D) = \pi_i(a_1, a_2) + 0 \left(= \pi_i(a_1, a_2) + \sum_{t=T-\tau+1}^T \pi_i(f^t(\tilde{h}_{t-1}(a_1, a_2))) \right),$$

where $\tilde{h}_{t-1}(a)$ is the date $t - 1$ history induced if $a = (a_1, a_2)$ is chosen at $h_{T-\tau-1}$ and invoking the functions f^t for $t = T - \tau + 1, \dots, T$. The unique NE of this game—with strategy sets equal to $\{C, D\}$ —is (D, D) . It follows that the unique SPNE of a finitely-repeated prisoners' dilemma is that each player i chooses *D after every history*: for $T < \infty$, $SPNE(G_{PD}^T) = \{(s_1^*, s_2^*)\}$, where, for $i = 1, 2$, s_i^* is given by $f_i^{t*}(h_{t-1}) = D$ for every $h_{t-1} \in H_{t-1}$ and for $t = 1, \dots, T$.

NE OUTCOME. I prove it by contraposition. Let $\tilde{s} \in S$ generate an outcome

$$\tilde{h}_T = h_0 \tilde{a}^1 \dots \tilde{a}^T \neq h_0 \underbrace{DD \dots DD}_{2T \text{ times}}.$$

I will show that $\tilde{s}$ is not an Nash equilibrium of NE_{PD}^T . Let τ be the *largest* t with $\tilde{a}^t \neq DD$, and, wlog, suppose that $\tilde{a}_1^\tau \neq D$. Now define $\hat{s}_1$ by

$$\hat{f}_1^t = \begin{cases} \tilde{f}_1^t & \text{if } t < \tau \\ D & \text{for every } h_{t-1} \in H_{t-1} \text{ if } t \geq \tau \end{cases}$$

The date- τ history that $(\hat{s}_1, \tilde{s}_2)$ generates is $\hat{h}_\tau = h_0 \tilde{a}^1 \dots \tilde{a}^{\tau-1} DD$, and, noting that $\pi_1(D, \tilde{a}_2^\tau) - \pi_1(C, \tilde{a}_2^\tau) = 1$, it follows that

$$u_1(\hat{s}_1, \tilde{s}_2) - u_i(\tilde{s}) = \underbrace{0}_{\tau-1 \text{ times}} + \underbrace{1}_{\text{date } \tau} + \sum_{t=\tau+1}^T \Delta_t > 0,$$

since $\Delta_t \in \{0, 3\}$ for $t > \tau$ (for $t > \tau$, either $\hat{a}^t = \tilde{a}^t = (D, D)$ or $\hat{a}^t = (D, C)$). It follows that $\tilde{s} \notin NE_{PD}^T$.